

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think> To solve , first calculate the distance to John 's initial destination . Then , calculate the distance he covers while trying to get home . The distance from home is the difference between these two distances . Distance = Speed $\times$ Time . For the first part , distance to initial destination = 60 mph $\times$ 3 hours = 180 miles . For the return trip , he spends 2 hours in traffic , so no distance covered in that time . For the next half-hour at 30 mph , distance covered = 30 mph $\times$ 0.5 hours = 15 miles . For the remaining time , he has $4 - 2 - 0.5 = 1.5$ hours left at 80 mph , so distance covered = 80 mph $\times$ 1.5 hours = 120 miles . Total distance covered trying to get home = 15 miles + 120 miles = 135 miles . So , distance from home = 180 miles - 135 miles = 45 miles . </think><answer> 45 </answer>

